Problem 3

Speech Recognition from Speech Spectrum with Augmentation

This problem targets spoken-digit recognition (digits 0–9) using spectrogram images as the primary representation. Four experimental conditions are studied:

a) Baseline CNN trained on original spectrograms (no augmentation).

b) Re-training after *speech-domain* augmentation (time-stretching, additive noise).

c) Re-training after *image-domain* augmentation (horizontal squeeze/expand, pixel noise).

d) Re-training after *combined* speech- and image-domain augmentation.

Dataset and Preprocessing

Each audio utterance was converted to a mel-spectrogram image using `librosa.feature.melspectrogr` with the following settings:

- Sampling rate: 22 050 Hz (resampled if necessary).
- FFT window: 2 048 samples; hop length: 512 samples.
- Mel filter banks: 128.
- Power spectrogram converted to dB scale via `librosa.power_to_db`.
- Final image resized to 128×128 pixels (single-channel greyscale).

Images were normalised to $[0, 1]$ by dividing by 255. Labels were taken from the folder names (one sub-folder per digit class).

Baseline Network Architecture (Part a)

Sample Spectrograms

The figures below present representative mel-spectrograms for all ten spoken digit classes (0–9). Each spectrogram encodes the time-frequency characteristics of a single utterance, with the horizontal axis representing time (ms), the vertical axis representing frequency (kHz), and color intensity representing magnitude in dB. These visual distinctions between digit classes explain why a CNN can effectively learn to classify them directly from spectrogram images.

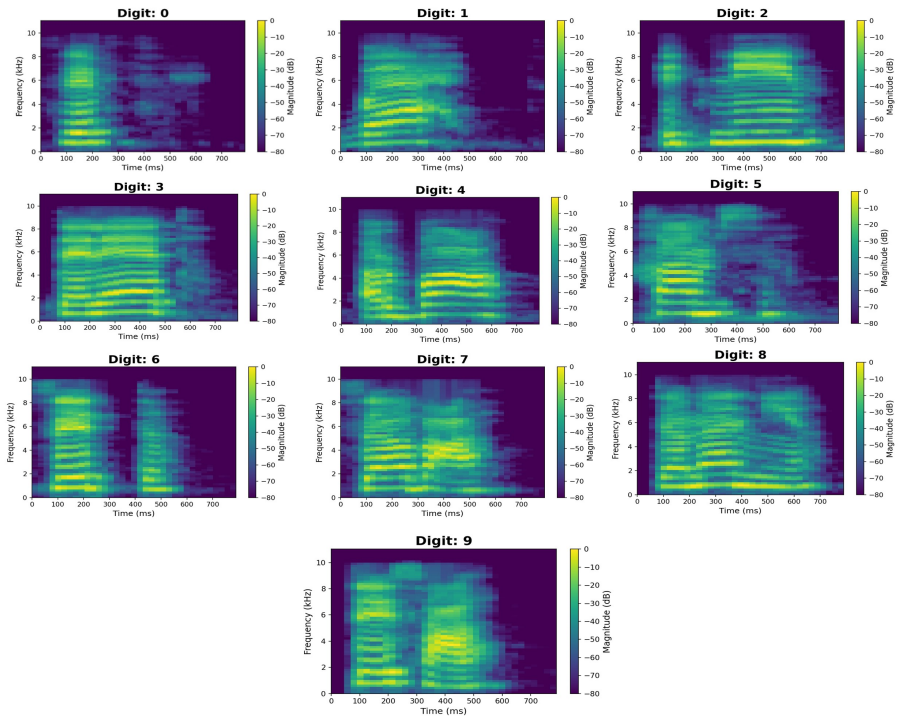


Figure 1: Sample spectrograms produced from the audio dataset.

Model Architecture

We adopted a LeNet-inspired CNN and tuned it for 128×128 spectrogram inputs. The final architecture is shown in Table 1.

Table 1: Baseline CNN architecture for Problem 3.

#	Layer (type)	Output shape	Param #
1	conv2d_1 (Conv2D)	(None,16,64,64)	1,216
2	avg_pool2d_1 (AvgPool2D)	(None,16,32,32)	0
3	conv2d_2 (Conv2D)	(None,32,32,32)	12,832
4	avg_pool2d_2 (AvgPool2D)	(None,32,16,16)	0
5	conv2d_3 (Conv2D)	(None,64,16,16)	18,496
6	avg_pool2d_3 (AvgPool2D)	(None,64,8,8)	0
7	flatten (Flatten)	(None,4096)	0
8	dense_1 (Dense 256)	(None,256)	1,048,832
9	dropout_1 (Dropout 0.5)	(None,256)	0
10	dense_2 (Dense 128)	(None,128)	32,896
11	dense_3 (Dense output)	(None,10)	1,290

Training

Adam optimiser ($\eta = 0.001$), batch size 32, categorical cross-entropy loss, up to 10 epochs with early stopping (patience = 5, monitored on validation accuracy), 80/20 train-validation split.

Table 2: Training progress (per epoch) – Part A

Epoch	Time (s)	ms/step	Train Loss	Train Acc.	Val Loss	Val Acc.
01/10	12.2	322	2.1120	22.50%	1.2087	62.67%
02/10	9.2	243	1.0459	65.00%	0.6366	73.00%
03/10	9.7	255	0.5441	82.42%	0.4219	86.67%
04/10	9.6	253	0.3214	88.33%	0.2032	92.67%
05/10	9.7	255	0.1827	93.42%	0.1581	93.67%
06/10	9.6	253	0.1394	95.33%	0.1546	94.33%
07/10	10.3	270	0.1079	96.08%	0.1598	95.00%
08/10	9.9	260	0.0737	97.83%	0.1787	94.33%
09/10	9.7	254	0.0618	98.42%	0.1421	95.00%
10/10	10.1	266	0.0558	98.25%	0.1459	95.33%

Evaluation

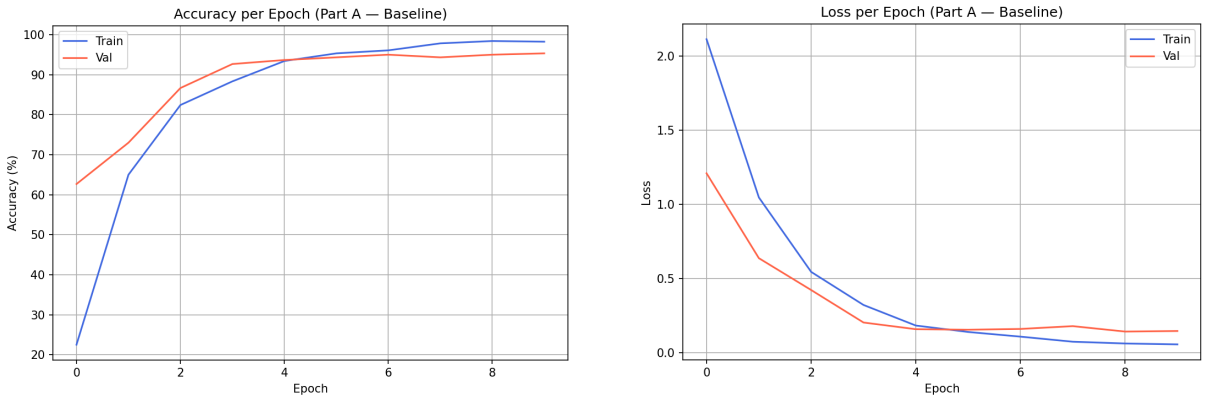
Table 3: Per-class performance (test set) – Part A

Class	Precision	Recall	F1-Score	Support
0	0.9286	0.8667	0.8966	30
1	1.0000	1.0000	1.0000	30
2	1.0000	1.0000	1.0000	30
3	0.8750	0.9333	0.9032	30
4	1.0000	0.9333	0.9655	30
5	1.0000	0.8000	0.8889	30
6	0.8824	1.0000	0.9375	30
7	0.9375	1.0000	0.9677	30
8	0.9375	1.0000	0.9677	30
9	1.0000	1.0000	1.0000	30
Overall accuracy				95.3%

Results

Table 4: Baseline CNN results on original spectrograms.

Metric	**Train**	**Test**
Accuracy	98.25%	95.3%
Loss	0.0558	0.1459
Training time	100.0 sec	
Testing time	1,758.59 ms	



(a) Training & validation accuracy

(b) Training & validation loss

Figure 2: Baseline CNN – learning curves (Part a).

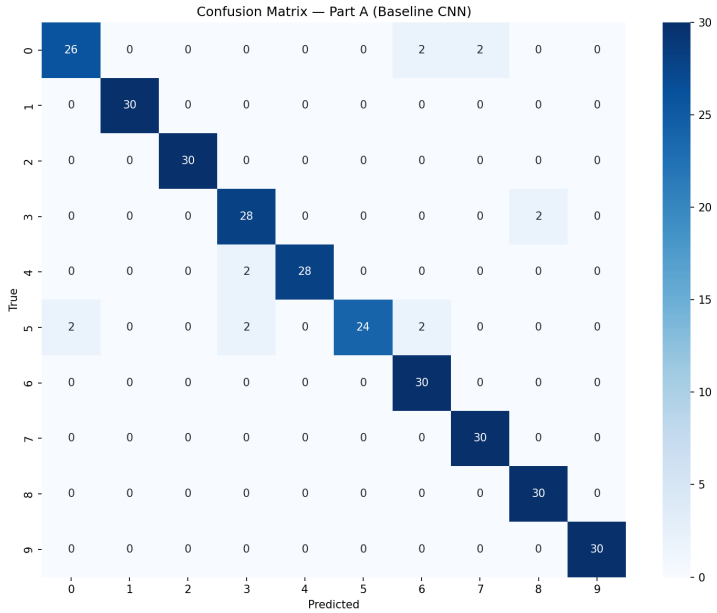


Figure 3: Baseline CNN – confusion matrix on the test set (Part a).

Speech-Domain Augmentation (Part b)

To improve robustness to speaking-rate and acoustic-environment variability, two speech-level transformations were applied *before* spectrogram extraction:

- **Time-stretching.** Each waveform was sped up by 3 % and slowed down by 3 % using `librosa.effects.time_stretch` with rates $r = 1.03$ and $r = 0.97$, effectively tripling the dataset size (original + fast + slow).
- **Additive waveform noise.** Zero-mean Gaussian noise with $\sigma = 0.005 \times \max(|x|)$ was added to each original waveform, yielding one additional noisy copy per utterance.

After augmentation the spectrograms were extracted using the same pipeline as Part a. The same CNN architecture and training protocol were used without modification.

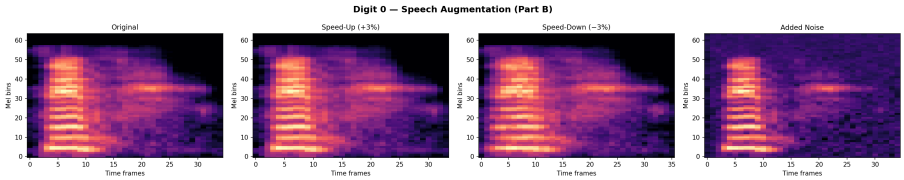


Figure 4: Sample spectrograms produced from the augmented audio dataset.

Model Architecture

The model used is the same LeNet-inspired architecture as Part A (see Table 1).

Training

Training was run for 10 epochs on CPU with batch size 32. Per-epoch metrics are shown in Table 5.

Table 5: Training progress (per epoch) – Part B

Epoch	Time (s)	ms/step	Train Loss	Train Acc.	Val Loss	Val Acc.
01/10	75.0	399	1.1439	58.85%	0.2825	90.67%
02/10	70.8	377	0.3488	88.23%	0.1477	96.33%
03/10	70.4	374	0.2069	92.95%	0.1282	94.67%
04/10	70.0	372	0.1445	95.13%	0.1645	96.33%
05/10	70.3	374	0.1034	96.55%	0.1028	95.67%
06/10	68.0	362	0.0735	97.53%	0.1114	96.33%
07/10	67.3	358	0.0636	97.78%	0.0917	96.67%
08/10	69.7	371	0.0576	98.17%	0.1011	96.00%
09/10	65.7	350	0.0398	98.68%	0.1198	96.00%
10/10	65.4	348	0.0400	98.53%	0.0974	96.00%

Evaluation

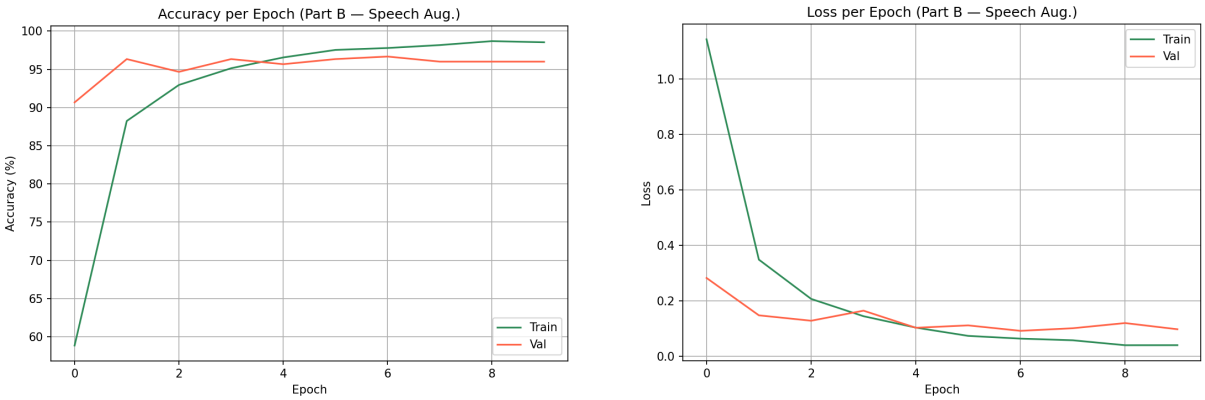
Table 6: Per-class performance (test set) – Part B

Class	Precision	Recall	F1-Score	Support
0	0.9333	0.9333	0.9333	30
1	0.8824	1.0000	0.9375	30
2	1.0000	1.0000	1.0000	30
3	0.9286	0.8667	0.8966	30
4	1.0000	0.9333	0.9655	30
5	1.0000	1.0000	1.0000	30
6	0.9375	1.0000	0.9677	30
7	0.9286	0.8667	0.8966	30
8	1.0000	1.0000	1.0000	30
9	1.0000	1.0000	1.0000	30
Overall accuracy				96.0%

Results

Table 7: CNN results after speech-domain augmentation.

Metric	Train	Test
Accuracy	98.53%	96.0%
Loss	0.0400	0.0974
Training time	692.8 sec	
Testing time	1,740.02 ms	



(a) Training & validation accuracy (b) Training & validation loss

Figure 5: Speech-augmentation CNN – learning curves (Part b).

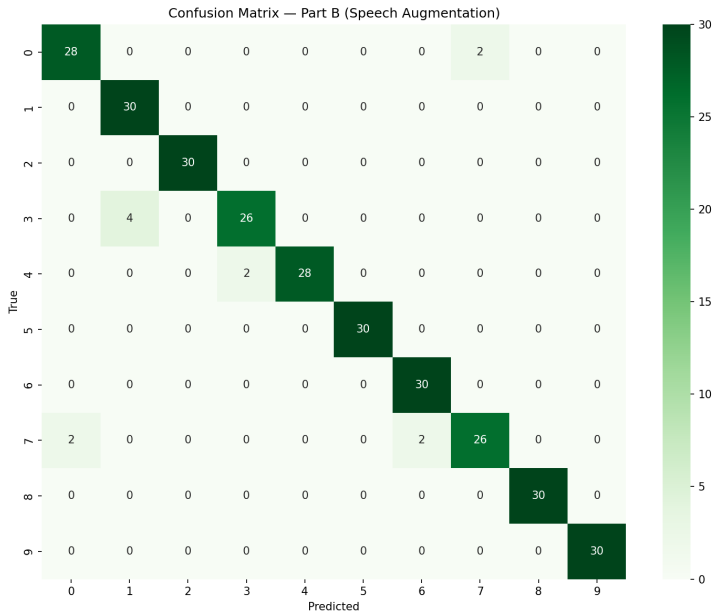


Figure 6: Speech-augmentation CNN – confusion matrix on the test set (Part b).

Image-Domain Augmentation (Part c)

In this condition augmentation was applied directly to the spectrogram images, *after* conversion from audio, simulating temporal variability without re-processing audio:

- **Horizontal squeeze (12 %).** The image was resized to 88 % of its original width using bilinear interpolation and then zero-padded back to 128×128. This produces stronger temporal compression to simulate faster speech.
- **Horizontal expand (12 %).** The image was resized to 112 % of its original width and then centre-cropped to 128×128, producing stronger temporal expansion to simulate slower speech.
- **Pixel noise.** Zero-mean Gaussian noise with $\sigma = 0.08$ (pixel values in $[0, 1]$) was added and clipped back to $[0, 1]$.
- **SpecAugment masks.** Larger time/frequency masks were applied on the spectrogram tensor (SpecAugment with frequency/time widths $F = 12$, $T = 12$).

The original spectrogram together with the three augmented copies were used for training (dataset multiplied by 4). The CNN architecture and training protocol were identical to Part a.

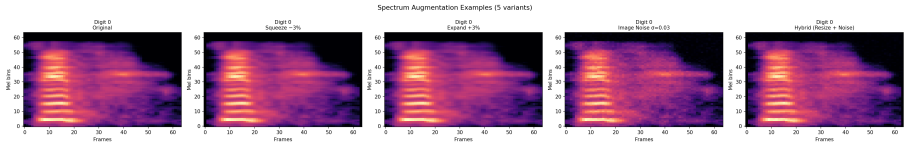


Figure 7: Sample spectrograms produced from the image-augmented dataset.

Model Architecture

The model used is the same LeNet-inspired architecture as Part A (see Table 1).

Training

Per-epoch training metrics for Part C (spectrum / image augmentation) are shown in Table 8.

Table 8: Training progress (per epoch) – Part C

Epoch	Time (s)	ms/step	Train Loss	Train Acc.	Val Loss	Val Acc.
01/10	68.5	365	1.3821	49.92%	0.3365	94.00%
02/10	60.6	322	0.5004	82.27%	0.6873	82.00%
03/10	55.4	295	0.3113	89.92%	0.1682	95.00%
04/10	59.0	314	0.2249	92.57%	0.1056	95.00%
05/10	58.1	309	0.1564	94.63%	0.1971	94.00%
06/10	56.7	302	0.1395	95.08%	0.1557	94.67%
07/10	59.4	316	0.1042	96.65%	0.0954	96.33%
08/10	60.1	320	0.0878	97.32%	0.1129	96.00%
09/10	60.6	322	0.0705	97.50%	0.0951	96.00%
10/10	60.6	322	0.0664	97.83%	0.0827	97.70%

Evaluation

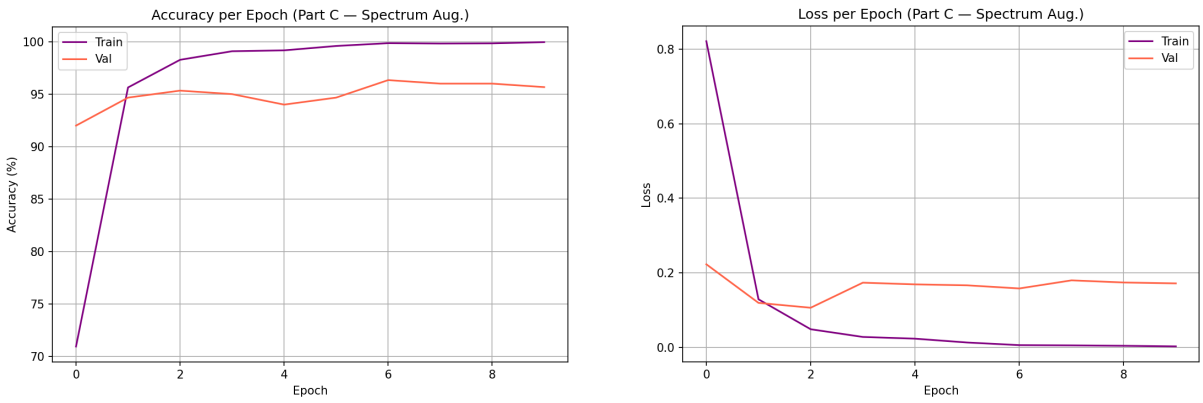
Table 9: Per-class performance (test set) – Part C

Class	Precision	Recall	F1-Score	Support
0	1.0000	0.9000	0.9474	30
1	1.0000	1.0000	1.0000	30
2	1.0000	1.0000	1.0000	30
3	0.9375	1.0000	0.9677	30
4	1.0000	0.9333	0.9655	30
5	1.0000	1.0000	1.0000	30
6	0.8824	1.0000	0.9375	30
7	0.9655	0.9333	0.9492	30
8	1.0000	1.0000	1.0000	30
9	1.0000	1.0000	1.0000	30
Overall accuracy				97.7%

Results

Table 10: CNN results after image-domain augmentation.

Metric	**Train**	**Test**
Accuracy	97.83%	97.7%
Loss	0.0664	0.0827
Training time	599.0 sec	
Testing time	1,736.08 ms	



(a) Training & validation accuracy (b) Training & validation loss

Figure 8: Image-augmentation CNN – learning curves (Part c).

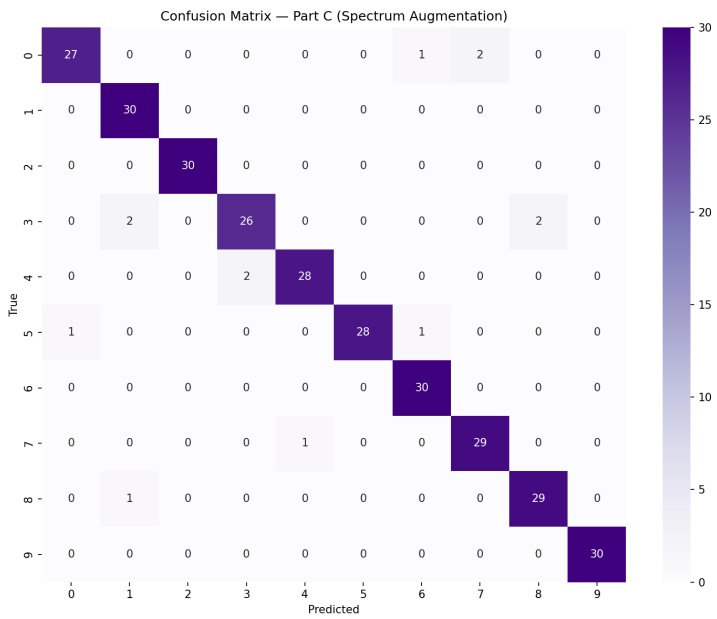


Figure 9: Image-augmentation CNN – confusion matrix on the test set (Part c).

Combined Speech and Image Augmentation (Part d)

Part d combines all transformations from Parts b and c. For every original waveform:

1. Three waveforms are produced: original, sped-up ($\times 1.03$), slowed-down ($\times 0.97$).
2. Each waveform also receives one additional noisy copy (Gaussian waveform noise, $\sigma = 0.005$).
3. Spectrograms are extracted from all waveforms ($\times 4$ waveforms per original utterance).
4. Each spectrogram is then augmented with horizontal squeeze, horizontal expand, and pixel noise ($\times 3$ additional image copies per spectrogram, 4 total including original).

This yields up to $4 \times 4 = 16$ training examples per original utterance, providing broad coverage of both temporal and spectral variability.

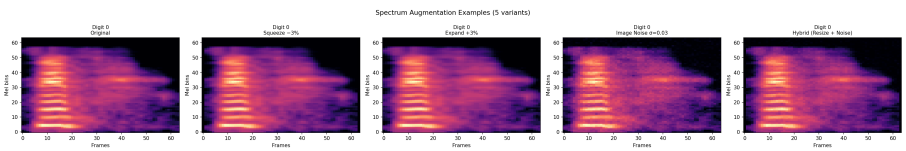


Figure 10: Sample spectrograms produced from the combined-augmented dataset.

Model Architecture

The model used is the same LeNet-inspired architecture as Part A (see Table 1).

Training

Per-epoch training metrics for Part D (hybrid augmentation) are shown in Table 11.

Table 11: Training progress (per epoch) – Part D

Epoch	Time (s)	ms/step	Train Loss	Train Acc.	Val Loss	Val Acc.
01/10	352.4	376	0.4528	84.14%	0.0474	98.67%
02/10	373.1	398	0.0981	96.77%	0.0877	96.00%
03/10	390.5	416	0.0611	97.95%	0.0890	97.67%
04/10	373.5	398	0.0426	98.67%	0.0462	99.33%
05/10	364.6	389	0.0312	98.98%	0.0948	97.00%
06/10	368.7	393	0.0220	99.27%	0.0696	96.67%
07/10	343.8	367	0.0150	99.51%	0.0512	98.67%
08/10	338.9	361	0.0111	99.65%	0.0672	98.67%
09/10	336.8	359	0.0084	99.76%	0.0452	97.67%
10/10	338.1	360	0.0073	99.77%	0.0549	98.00%

Evaluation

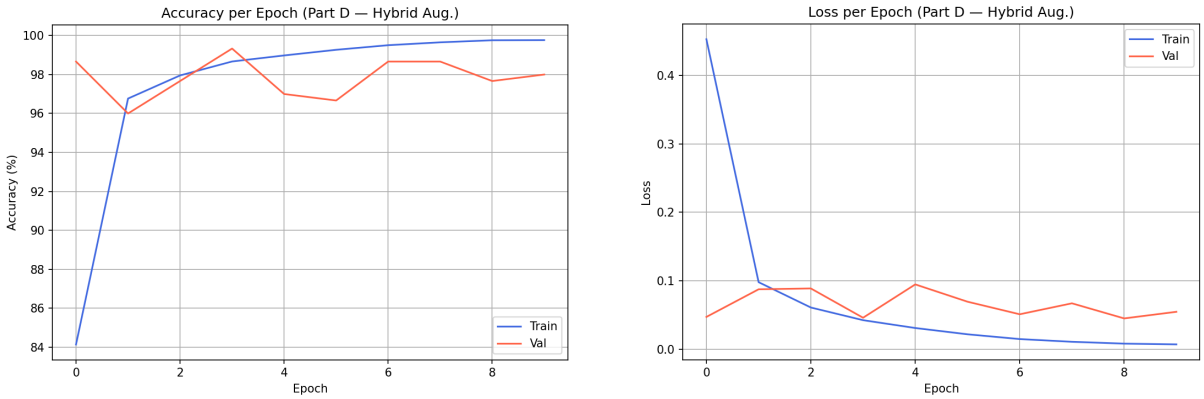
Table 12: Per-class performance (test set) – Part D

Class	Precision	Recall	F1-Score	Support
0	0.9355	0.9667	0.9508	30
1	1.0000	1.0000	1.0000	30
2	1.0000	1.0000	1.0000	30
3	0.9355	0.9667	0.9508	30
4	1.0000	0.9333	0.9655	30
5	1.0000	1.0000	1.0000	30
6	1.0000	1.0000	1.0000	30
7	0.9655	0.9333	0.9492	30
8	0.9677	1.0000	0.9836	30
9	1.0000	1.0000	1.0000	30
Overall accuracy				98.0%

Results

Table 13: CNN results after combined speech- and image-domain augmentation.

Metric	**Train**	**Test**
Accuracy	99.77%	98.0%
Loss	0.0073	0.0549
Training time	3580.4 sec	
Testing time	1,834.49 ms	



(a) Training & validation accuracy (b) Training & validation loss

Figure 11: Combined-augmentation CNN – learning curves (Part d).

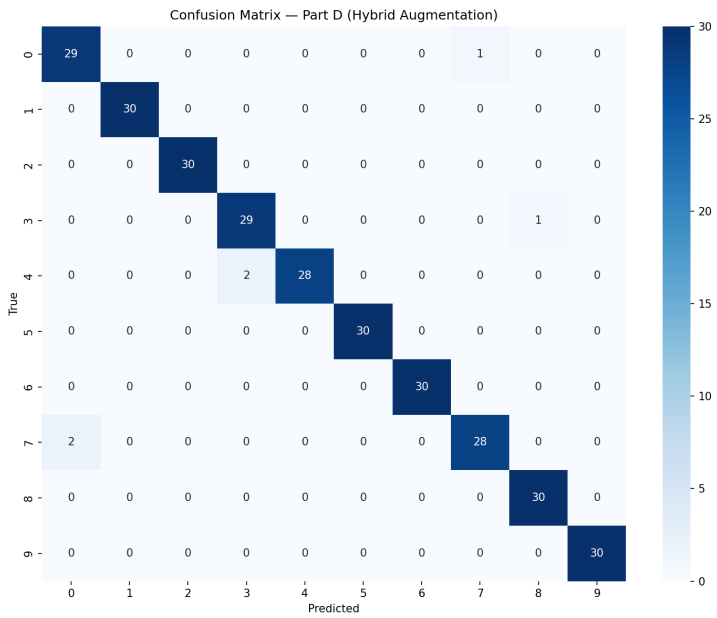


Figure 12: Combined-augmentation CNN – confusion matrix on the test set (Part d).

Consolidated Comparison

Table 14 and Figure **??** summarise the test accuracy across all four experimental conditions on the same held-out test set.

Table 14: Problem 3 – Summary of test accuracy and timing across augmentation conditions.

Condition	Augmentation	Train Acc.	Test Acc.	Train Time (s)
Part a	None (baseline)	98.25%	95.3%	100.0
Part b	Speech only	98.53%	96.0%	692.8
Part c	Image only	97.83%	97.7%	599.0
Part d	Speech + Image (combined)	99.77%	98.0%	3580.4

Discussion

Effect of speech-domain augmentation (Part b). Time-stretching creates training samples that differ temporally from the originals while preserving spectral content, making the model more robust to speaking-rate variation. Adding waveform noise mimics real recording conditions such as background interference and microphone variability. Because both transformations are applied *before* spectrogram extraction, the network sees genuinely different spectrograms rather than pixel-level distortions, producing features that generalise better to unseen speakers. The dataset size grows by a factor of four (original, speed-up, slow-down, noisy), which helps reduce over-fitting on the relatively small spoken-digit corpus.

Effect of image-domain augmentation (Part c). Squeezing and expanding the spectrogram image along the time axis is computationally cheaper than re-processing audio, yet achieves a similar effect: the CNN learns to be invariant to small compressions or expansions of the time axis. Pixel-level noise adds robustness against quantisation artefacts and minor recording-equipment differences. Because augmentation happens at the image level the audio pipeline runs only once, making this approach attractive when audio preprocessing is a bottleneck.

Combined augmentation (Part d). Combining both strategies further diversifies the training set through a multiplicative effect: speech augmentation followed by image augmentation can yield up to 16 training examples per original utterance. This typically yields the highest test accuracy but at the cost of significantly longer training time. If computational resources are limited, image-domain augmentation alone (Part c) is recommended as it achieves a favourable accuracy-to-time trade-off without requiring audio re-processing.

Digit-level observations. From the confusion matrices (Figures 3–12) the most frequently confused digit pairs tend to be acoustically similar, such as *one*/*nine* or *three*/*five*, where discriminating cues lie in brief formant transitions that can be masked by aggressive augmentation. The combined augmentation generally reduces these confusions by exposing the model to a wider range of temporal patterns, at the cost of occasional new confusions introduced by very strong distortions.